

AMPLIFICATION & DSP

DAP-8

8-CHANNEL DSP POWER AMPLIFIER

Eight channels of class-D amplification and full system DSP in 2U, built to run two complete D&A loudspeakers.

8 CHANNELS

BRIDGEABLE IN PAIRS

2000 W

PER CHANNEL @ 4 Ω

OVERVIEW

DAP-8 is D&A Innovations' custom eight-channel amplifier platform. Each channel delivers up to 2000 W into 4 Ω, with pairs bridgeable to 3000 W into 8 Ω for subwoofers. Two signal inputs each feed four output channels, so one DAP-8 runs two 4-way cabinets. Crossovers, FIR and IIR equalisation, delay and limiters run inside the amplifier, so the approved presets for D&A loudspeakers load directly without an external processor. A network link carries digital audio and control in redundant, daisy-chained layouts.

OUTPUT STAGE

POWER PER CHANNEL @ 16 Ω	600 W
POWER PER CHANNEL @ 8 Ω	1000 W
POWER PER CHANNEL @ 4 Ω	2000 W
BRIDGED @ 16 Ω / 8 Ω	2000 W / 3000 W
MAX OUTPUT VOLTAGE / CURRENT	150 V peak / 54 A peak
FREQUENCY RESPONSE	20 Hz - 20 kHz (+0 / -1 dB, 1 W @ 8 Ω)
S/N RATIO / THD+N	115 dB (A) / < 0.09% (typ. < 0.05%)
DAMPING FACTOR	> 500 @ 100 Hz

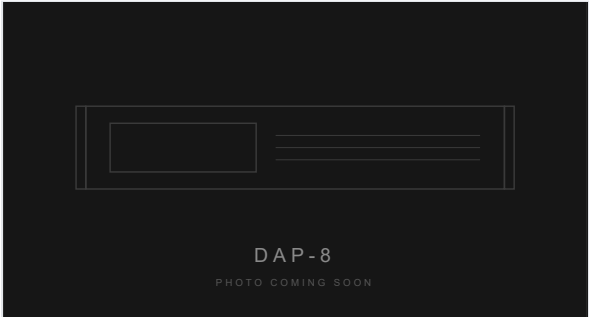
Per-channel power is the maximum for a single channel.

POWER & MECHANICAL

AC MAINS	100 - 240 V, 50/60 Hz, universal PFC supply
OPERATING VOLTAGE	90 - 264 V
POWER FACTOR	> 0.90 @ 4 Ω full power
CONSUMPTION	1200 VA @ 1/8 max power, 8 Ω
COOLING	Variable-speed fan, temperature controlled
OPERATING TEMPERATURE	0 - 35 °C, 0 - 90% RH non-condensing
FORMAT	2U, 19-inch (483 mm) rack width, standard amplifier depth

SIGNAL PROCESSING

40-BIT FLOAT PROCESSING 330 MHz SHARC, 4.02 ms latency 24-bit Tandem AD/DA, 32 - 192 kHz in	FIR + IIR FILTERS & CROSSOVERS Custom FIR, parametric IIR; LR, Butterworth, Bessel 6-48 dB/oct	PROTECTION LIMITERS & DELAY TruePower, RMS, peak; DampingControl 4 s input + 100 ms output delay
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INPUTS & OUTPUTS:

- Analog: 2x XLR + 2x XLR link thru
AES3: 2x XLR
Network: 4x RJ45, AES3 audio and control, redundant daisy-chain
- Outputs: 4x speakON NL4 (2 channels each)
Option: 2x NL8 (4 channels each)
Control: Ethernet, remote monitoring, up to 10 presets per input

OPERATIONAL REQUIREMENTS

- System presets:** Load the approved D&A preset for each connected loudspeaker before operation. Stated loudspeaker specifications are only valid with the matching preset and limiter settings.
- Mains:** Use a dedicated circuit rated for the full load. Keep front and rear vents clear.